

(5)

Z(5th Sm.)-Computer Sc.-G/DSE-A-1,
DSE-A-2 & DSE-A-3/CBCS

Paper : DSE-A-3
(Computer Graphics)

Full Marks : 50

The figures in the margin indicate full marks.

Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions :

2×5

- (a) What is aspect ratio?
- (b) Differentiate between monochrome and color monitor.
- (c) Write down 2-D shearing matrix about X-axis and about Y-axis.
- (d) What is frame buffer?
- (e) What do you mean by scan conversion?
- (f) What do you mean by inverse transformation?
- (g) What do you understand by interior and exterior clipping?
- (h) What do you mean by animation?

2. Discuss Bresenham's algorithm for line drawing. Explain why this algorithm is considered to be preferred over DDA and midpoint for line drawing.

5+5

3. (a) Why is translation matrix in 2D needed to be converted in homogeneous form? Explain in detail.
(b) Mention the features of a CRT monitor.
(c) What do you mean by scaling? Explain its types.

4+3+3

4. (a) What is 2D rotation? Derive the matrix for each type of rotation.

- (b) Consider the square A(1,0), B(0,0), C(1,0), D(1,1). Rotate the square ABCD by 45° clockwise about A(1,0).

(2+3)+5

5. (a) Briefly explain the 2D reflection method.

- (b) Write 2D transformation matrix to find reflection of a point P(10,20) about the line $y = x + 2$.

5+5

6. (a) What is the difference between a window and a viewport?

- (b) Explain the Cohen Sutherland line clipping algorithm.

(c) Given a clipping window P(0,0), Q(30,0), R(30,20), S(0,20). Determine the visible portion of the line A(10,30) and B(40,0).

2+4+4

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7. (a) Discuss Sutherland-Hodgman polygon clipping algorithm.
(b) Differentiate between parallel projection and perspective projection.
(c) What is affine transformation?
8. (a) Show that the composition of two successive rotations about origin is additive.
(b) Briefly describe the steps of morphing.

5+3+2

5+5